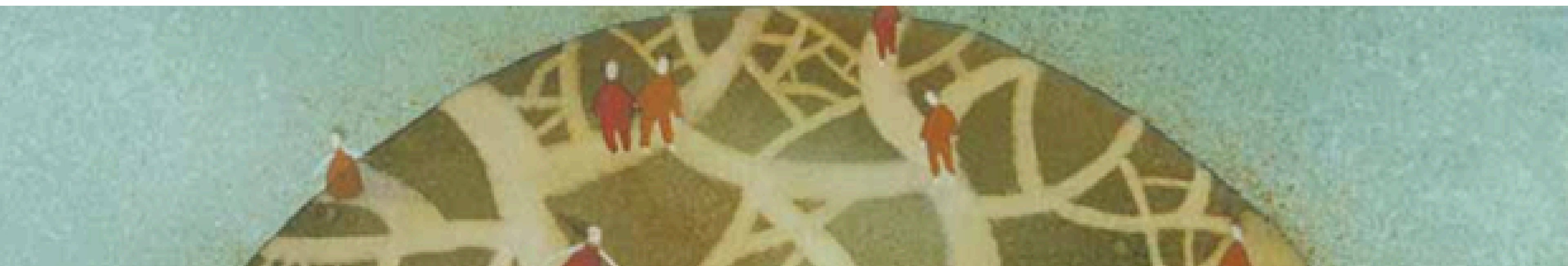


CHAPTER 1

THE NATURE OF PSYCHOLOGY



Topics

The Scope of Psychology 5

Interim Summary 7

Critical Thinking Questions 7

The Historical Origins of Psychology 7

Nature–nurture debate 7

The beginnings of scientific psychology 8

Structuralism and functionalism 8

Behaviorism 9

Gestalt psychology 9

Psychoanalysis 10

Later developments in twentieth-century psychology 11

Interim Summary 11

Critical Thinking Questions 11

Contemporary Psychological Perspectives 12

The biological perspective 12

The behavioral perspective 13

The cognitive perspective 13

The psychoanalytic perspective 14

The subjectivist perspective 15

Relationships between psychological
and biological perspectives 15

Major subfields of psychology 16

Interim Summary 17

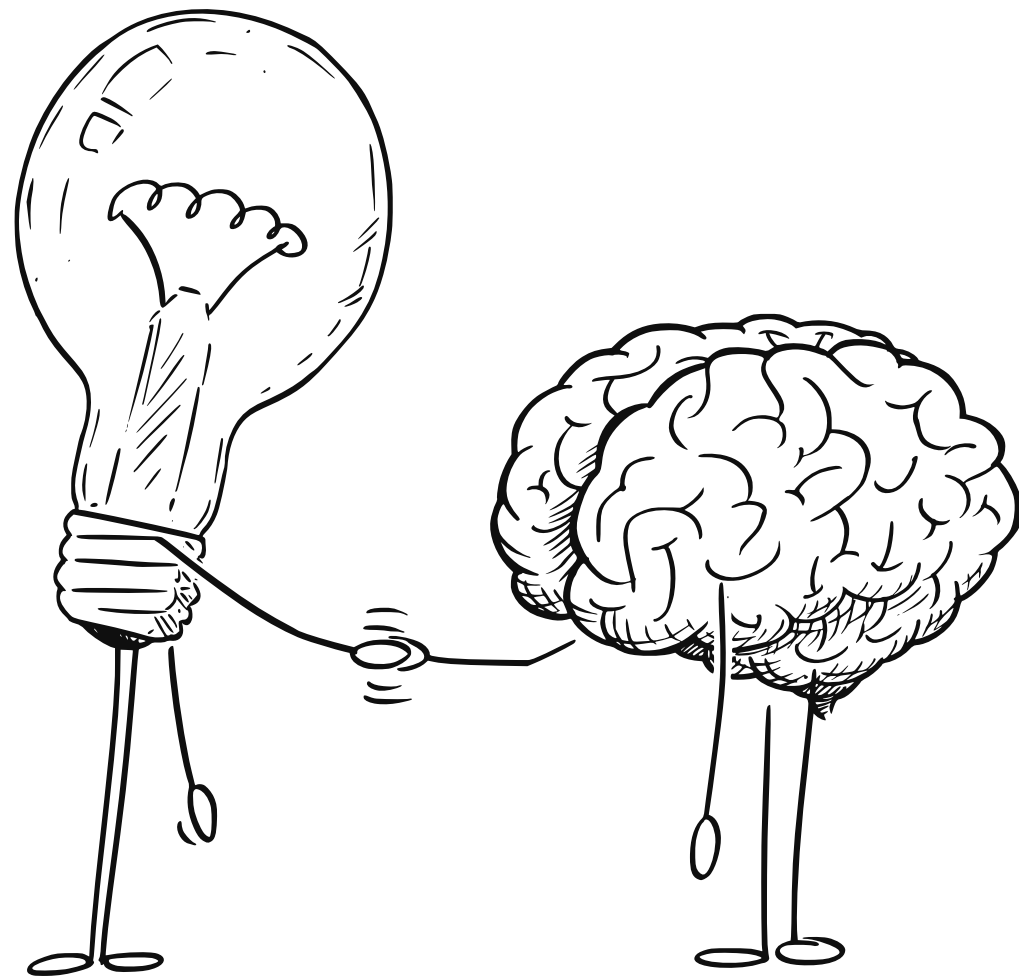
Critical Thinking Questions 17

The Scope of Psychology

Psychology can be defined as the scientific study of behavior and mental processes.

An astonishing variety of topics fit this definition, as can be seen in the brief examples presented next.

Brain damage and face recognition



It is no surprise that when people suffer brain damage, their behavior is affected. What is surprising is that damage in a specific part of the brain may change a person's behavior in one way but not in any other ways. In some cases, for example, people are unable to recognize familiar faces as a result of damage to a particular region on the right side of the brain – yet they can do just about everything else normally, a condition called **prosopagnosia**. A famous example of this condition was described by neurologist Oliver Sacks (1985) in his book *The Man Who Mistook His Wife for a Hat*. In another case, a man with prosopagnosia complained to a waiter that someone was staring at him, only to be informed that he was looking in a mirror! Such cases tell us a lot about the way the normal brain works. They indicate that some psychological functions – like face recognition – are localized in particular parts of the brain.

Attributing traits to people

Suppose that in a crowded department store a person soliciting for a charity approaches a customer and implores her to make a contribution. The woman donates a small sum to the charity. Would you think the woman was generous, or would you think she had been pressured into making the donation because so many people were watching her? Experiments designed to study situations like this have shown that most people consider the woman generous, even though the situational pressures were so great that just about everybody would behave similarly. When explaining the behavior of others, people tend to overestimate the causal effect of personality traits and underestimate those of situational factors – a mistake social psychologists call the **fundamental attribution error** (see Figure 1.1). If we contrast the fundamental attribution error with the overjustification effect (discussed in the context of earning pizzas for reading), we begin to see some important distinctions between how we judge others and how we judge ourselves. When making sense of our own behavior, we often overestimate – not underestimate – situational causes.



Figure 1.1 Trait Attribution. *In deciding whether another person's substantial donation to charity is caused by the giver's traits or by the situation, we are biased toward believing that a trait was the critical factor. This illustrates the fundamental attribution error.*

Obesity

More than a quarter of U.S. adults are **obese**; their weight is 30 percent or more above the level that would be appropriate for their body structure and height. Obesity is dangerous. It increases vulnerability to diabetes, high blood pressure, and heart disease. Psychologists are interested in what factors lead people to eat too much. One factor seems to be a history of deprivation. If rats are first deprived of food, then allowed to eat until they return to their normal weight, and finally allowed to eat as much as they want, they eat more than rats that have no history of deprivation.



Effects of media violence on children's aggression

The question of whether watching violence on television causes children to be more aggressive has long been controversial. Although many observers believe that televised violence affects children's behavior, others suggest that watching violence has a **cathartic effect**. It may actually reduce aggression by allowing children to express it vicariously and 'get it out of their system'. But research evidence does not support the cathartic effect view. In one experiment, one group of children watched violent cartoons while another group watched nonviolent cartoons for the same amount of time. Children who watched violent cartoons became more aggressive in their interactions with peers, but the children who viewed nonviolent cartoons showed no change in aggressive behavior. These effects can persist over time: The more violent programs a boy watches at age 9, the more aggressive he is likely to be at age 19 (see Figure 1.3).

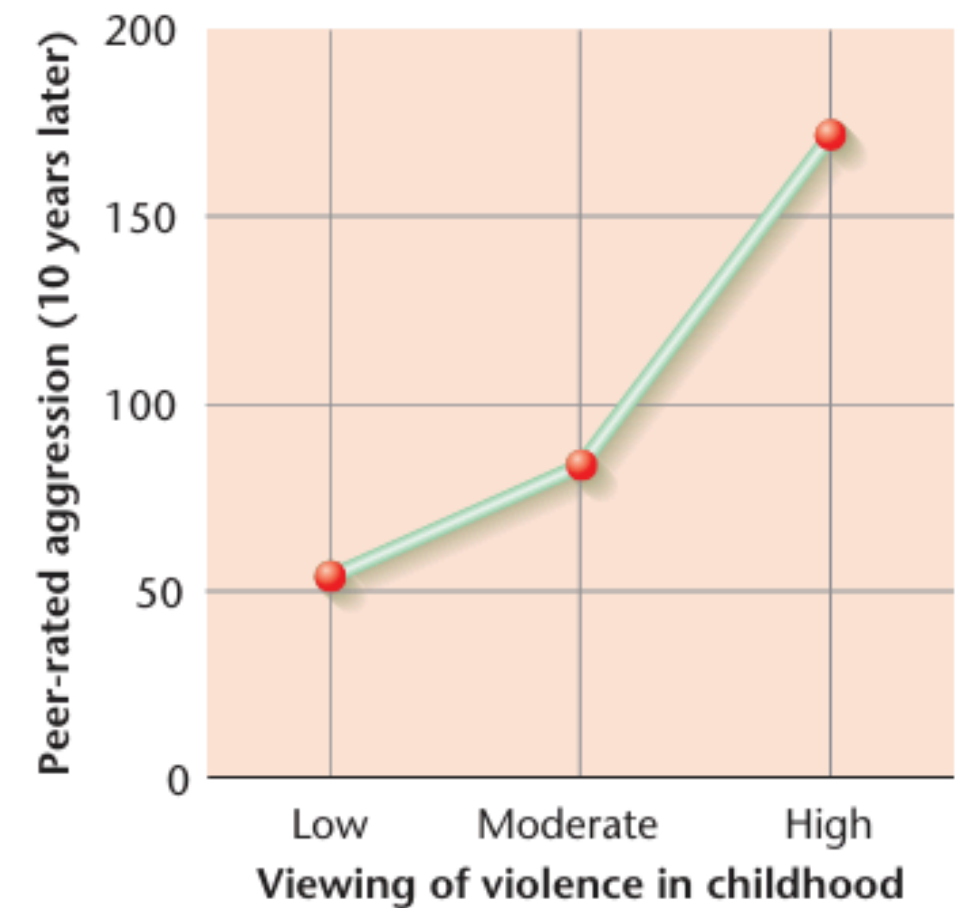


Figure 1.3 The Relationship Between Childhood Viewing of Violent Television and Adult Aggression. A classic study shows that preference for viewing violent TV programs by boys at age 9 is related to aggressive behavior as rated by peers at age 19. (L. Eron, et al. (1972) 'Does Television Violence Cause Aggression?' *American Psychologist*, 27:253–262. Copyright © 1972 by The American Psychological Association. Adapted by permission.)

The Scope of Psychology

INTERIM SUMMARY

- Psychology touches on many aspects of our lives and influences laws and public policy.
- To evaluate new claims made about psychology, you need to know (1) what psychological facts are already firmly established and (2) the standards for scientific evidence.
- Psychology is the scientific study of behavior and mental processes.
- The scope of psychology is broad, covering topics such as face recognition, social judgments, memory, obesity, violence, and many more.

The Historical Origins of Psychology

Nature-Nurture Debate

This **nature–nurture** debate centers on the question of whether human capabilities are inborn or acquired through experience.

The **nature** view holds that human beings enter the world with an inborn store of knowledge and understanding of reality. Early philosophers believed that this knowledge and understanding could be accessed through careful reasoning and introspection. In the seventeenth century, Descartes supported the nature view by arguing that some ideas (such as God, the self, geometric axioms, perfection, and infinity) are innate. Descartes is also notable for his conception of the body as a machine that can be studied much as other machines are studied. This is the root of modern information-processing perspectives on the mind, discussed later in this chapter.

The **nurture view** holds that knowledge is acquired through experiences and interactions with the world. Although some of the early Greek philosophers had this opinion, it is most strongly associated with the seventeenth-century English philosopher John Locke. According to Locke, at birth the human mind is a tabula rasa, a blank slate on which experience 'writes' knowledge and understanding as the individual matures. This perspective gave birth to **associationist psychology**. Associationists denied that there were inborn ideas or capabilities. Instead, they argued that the mind is filled with ideas that enter by way of the senses and then become associated through principles such as similarity and contrast. Current research on memory and learning is related to early association theory.

The Beginnings of Scientific Psychology

Wilhelm Wundt relied on introspection to study mental processes. **Introspection** refers to observing and recording the nature of one's own perceptions, thoughts, and feelings. Examples of introspections include people's reports of how heavy they perceive an object to be and how bright a flash of light seems to be. The introspective method was inherited from philosophy, but Wundt added a new dimension to the concept. Pure self-observation was not sufficient; it had to be supplemented by experiments. Wundt's experiments systematically varied some physical dimension of a stimulus, such as its intensity, and used the introspective method to determine how these physical changes modified the participant's conscious experience of the stimulus.

The reliance on introspection, particularly for very rapid mental events, proved unworkable. Even after extensive training, different people produced very different introspections about simple sensory experiences, and few conclusions could be drawn from these differences. As a result, introspection is not a central part of the current cognitive perspective. And, as we will see, some psychologists' reactions to introspection played a role in the development of other modern perspectives.

Structuralism and Functionalism

E. B. Titchener, a Cornell University psychologist who had been trained by Wundt. Titchener introduced the term **structuralism**– the analysis of mental structures– to describe this branch of psychology.

But some psychologists opposed the purely analytic nature of structuralism. William James, a distinguished psychologist at Harvard University, felt that analyzing the elements of consciousness was less important than understanding its fluid, personal nature. His approach was named **functionalism**, studying how the mind works to enable an organism to adapt to and function in its environment.

Nineteenth-century psychologists' interest in adaptation stemmed from the publication of Charles Darwin's theory of evolution. Some argued that consciousness had evolved only because it served some purpose in guiding the individual's activities. To find out how an organism adapts to its environment, functionalists said that psychologists must observe actual behavior. However, both structuralists and functionalists still regarded psychology as the science of conscious experience.



Structuralism and functionalism played important roles in the early development of twentieth-century psychology. Because each viewpoint provided a systematic approach to the field, they were considered competing schools of psychology. By 1920, however, both were being displaced by three newer schools: **behaviorism**, **Gestalt psychology**, and **psychoanalysis**. Of the three, behaviorism had the greatest influence on scientific psychology in North America.

Behaviorism

Behaviorism is a theory of learning based on the idea that all behaviors are acquired through conditioning, and conditioning occurs through interaction with the environment.

John B. Watson, reacted against the view that conscious experience was the province of psychology. Watson made no assertions about consciousness when he studied the behavior of animals and infants.

For psychology to be a science, Watson believed, psychological data must be open to public inspection like the data of any other science. Behavior is public; consciousness is private. Science should deal only with public facts. Because psychologists were growing impatient with introspection, the new behaviorism caught on rapidly, and many younger psychologists in the United States called themselves 'behaviorists'.

Watson, and others ascribing to behaviorism, argued that **nearly all behavior is a result of conditioning and the environment shapes behavior by reinforcing specific habits**. For example, giving children cookies to stop them from whining reinforces (rewards) the habit of whining. The **conditioned response** was viewed as the smallest unit of behavior, from which more complicated behaviors could be created. All types of complex behavior patterns coming from special training or education were regarded as nothing more than an interlinked fabric of conditioned responses. Behaviorists tended to discuss psychological phenomena in terms of stimuli and responses, giving rise to the term stimulus–response (S–R) psychology.

Note, how ever, that S-R psychology itself is not a theory or perspective but a set of terms that can be used to communicate psychological information. S–R terminology is still sometimes used in psychology today.



Gestalt Psychology

About 1912, at the same time that behaviorism was catching on in the United States, Gestalt psychology was appearing in Germany. Gestalt is a German word meaning 'form' or 'configuration', which referred to the approach taken by **Max Wertheimer** and his colleagues **Kurt Koffka** and **Wolfgang Köhler**, all of whom eventually emigrated to the United States.

The Gestalt psychologists' primary interest was perception, and they believed that perceptual experiences depend on the patterns formed by stimuli and on the organization of experience. What we actually see is related to the background against which an object appears, as well as to other aspects of the overall pattern of stimulation (see Chapter 5). The whole is different from the sum of its parts, because the whole depends on the relationships among the parts. For example, when we look at Figure 1.4, we see it as a single large triangle as a single form or Gestalt— rather than as three small angles.

The Gestalt psychologists also influenced key founders of modern social psychology– including Kurt Lewin, Solomon Asch, and Fritz Heider– who expanded on Gestalt principles to understand interpersonal phenomena (Jones, 1998). For instance, Asch (1946) extended the Gestalt notion that people see wholes rather than isolated parts from the simple case of object perception to the more complex case of person perception (Taylor, 1998). Plus, they saw the process of imposing meaning and structure on incoming stimuli as automatic and outside conscious awareness, a Gestalt view that continues to infuse contemporary research on social cognition to this day (see Chapter 18; Moskowitz, Skurnik, & Galinsky, 1999).

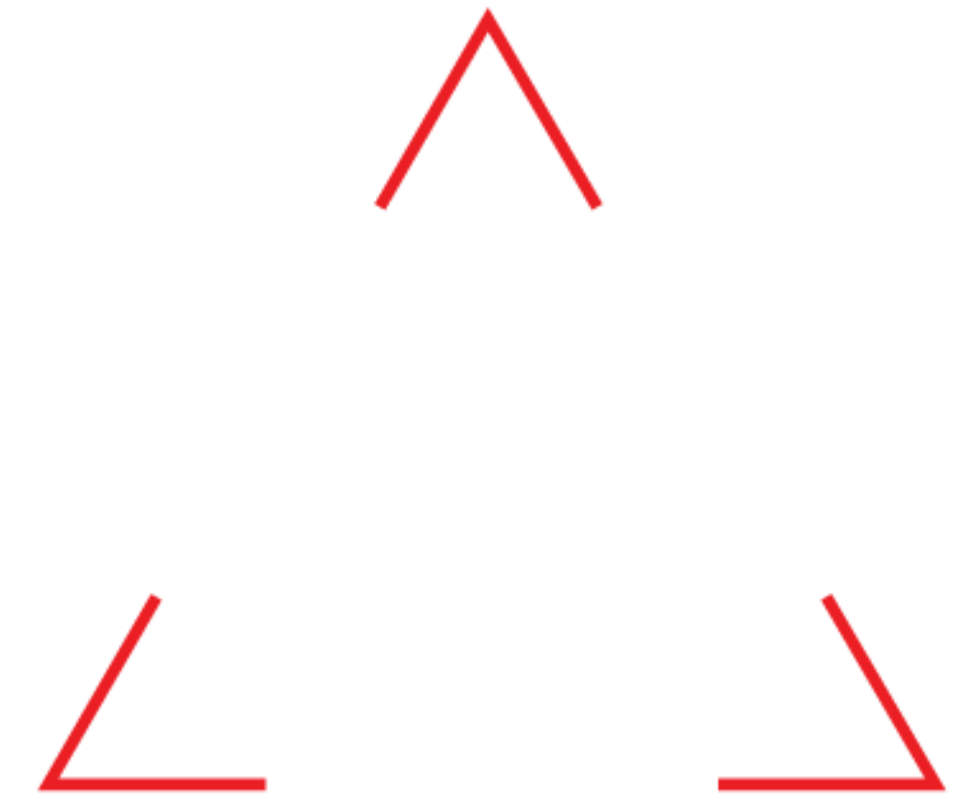


Figure 1.4 A Gestalt Image. *When we look at the three angles of an equilateral triangle, we see a single large triangle rather than three small angles.*

Psychoanalysis

Psychoanalysis is both a theory of personality and a method of psychotherapy originated by Sigmund Freud around the turn of the twentieth century.

At the center of Freud's theory is the concept of the **unconscious**– the thoughts, attitudes, impulses, wishes, motivations, and emotions of which we are unaware. Freud believed that childhood's unacceptable (forbidden or punished) wishes are driven out of conscious awareness and become part of the unconscious, where they continue to influence our thoughts, feelings, and actions. Unconscious thoughts are expressed in dreams, slips of the tongue, and physical mannerisms. During therapy with patients, Freud used the method of **free association**, in which the patient was instructed to say whatever comes to mind as a way of bringing unconscious wishes into awareness. The analysis of dreams served the same purpose.

In classical Freudian theory, the motivations behind unconscious wishes almost always involved sex or aggression. For this reason, Freud's theory was not widely accepted when it was first proposed. Contemporary psychologists do not accept Freud's theory in its entirety, but they tend to agree that people's ideas, goals, and motives can at times operate outside conscious awareness.

Later Developments in Twentieth-Century Psychology

- Information-processing theory
- Psycholinguistics
- Neuropsychology

The Historical Origins of Psychology

INTERIM SUMMARY

- The roots of psychology can be traced to the 4th and 5th centuries B.C. One of the earliest debates about human psychology focused on the question of whether human capabilities are inborn or acquired through experience (the nature–nurture debate).
- Scientific psychology was born in the late nineteenth century with the idea that mind and behavior could be the subject of scientific analysis. The first experimental laboratory in psychology was established by Wilhelm Wundt at the University of Leipzig in 1879.
- Among the early ‘schools’ of psychology in the twentieth century were structuralism, functionalism, behaviorism, Gestalt psychology, and psychoanalysis.
- Later developments in twentieth-century psychology included information-processing theory, psycholinguistics, and neuropsychology.

Five perspectives within psychology

Biological perspective	An orientation toward understanding the neurobiological processes that underlie behavior and mental processes.
Behavioral perspective	An orientation toward understanding observable behavior in terms of conditioning and reinforcement.
Cognitive perspective	An orientation toward understanding mental processes such as perceiving, remembering, reasoning, deciding, and problem solving and their relationship to behavior.
Psychoanalytic perspective	An orientation toward understanding behavior in terms of unconscious motives stemming from sexual and aggressive impulses.
Subjectivist perspective	An orientation toward understanding behavior and mental processes in terms of the subjective realities people actively construct.

Contemporary Psychological Perspectives

Contemporary Psychological Perspectives

Basically, it is an approach, a way of looking at topics within psychology.

Suppose that, following an insult, you punch someone in the face. From a **biological perspective**, we can describe this act as involving certain brain areas and as the firing of nerves that activate the muscles that move your arm. From a **behavioral perspective**, we can describe the act without reference to anything within your body; rather, the insult is a stimulus to which you respond by punching, a learned response that has been rewarded in the past. A **cognitive perspective** on this action would focus on the mental processes involved in producing the behavior, and we might explain your punch in terms of your goals and plans: Your goal is to defend your honor, and aggressive behavior is part of your plan for achieving that goal. From a **psychoanalytic perspective**, your action could be described as an expression of an unconscious aggressive instinct. And finally, from a **subjectivist perspective**, your aggressive act can be understood as a reaction to interpreting the person's utterance as a personal insult.₂₆

Keep in mind that these approaches need not be mutually exclusive; rather, they may focus on different aspects of the same complex phenomenon. In fact, understanding many psychological topics requires an **eclectic approach** that spans multiple perspectives.

1. The Biological Perspective

Research from the biological perspective seeks to specify the neurobiological processes that underlie behavior and mental processes.

- The biological approach to depression, for example, tries to understand this disorder in terms of abnormal changes in levels of neurotransmitters, which are chemicals produced in the brain that make communication between nerve cells possible.
- The biological perspective has also assisted in the study of memory. It emphasizes the importance of certain brain structures, including the hippocampus, which is involved in consolidating memories. Childhood amnesia may be partly due to an immature hippocampus, a structure that is not fully developed until a year or two after birth.

2. The Behavioral Perspective

- The behavioral perspective focuses on observable stimuli and responses and regards nearly all behavior as a result of conditioning and reinforcement.
- Behaviorism is like saying that we learn how to do things because of what happens after we do them.
- Imagine your teacher gives you a sticker every time you answer a question correctly in class. Because you like getting stickers, you start trying your best to answer more questions correctly. You've learned to behave in a way that gets you a reward! That's behaviorism in action!

3. The Cognitive Perspective

- The cognitive perspective assumes (1) Only by studying mental processes can we fully understand what organisms do, and (2) we can study mental processes in an objective fashion by focusing on specific behaviors (just as behaviorists do) but interpreting them in terms of underlying mental processes such as, perceiving, remembering, reasoning, deciding, and problem solving.
- In making these interpretations, cognitive psychologists have often relied on an analogy between the mind and a computer. Incoming information is processed in various ways: It is selected, compared, and combined with other information already in memory, transformed, rearranged, and so on.

4. The Psychoanalytic Perspective

The basic assumption of the psychoanalytic perspective is that behavior stems from unconscious processes, meaning beliefs, fears, and desires that a person is unaware of but that nonetheless influence behavior.

Key beliefs of this approach:

- Much of our behavior is caused by unconscious thoughts and feelings
- Childhood experiences have a strong impact on our adult personality
- Our behavior and feelings are influenced by internal conflicts we're not aware of
- Mental problems come from unresolved issues, often from childhood

Imagine you forget an important dentist appointment that you made weeks ago. While you might say, "I just forgot," a psychoanalyst would look deeper. They might suggest that you unconsciously wanted to avoid the appointment because you fear dental work. Your mind "helped" you forget because of this hidden fear, even though you weren't consciously aware of trying to avoid it.

The Psychoanalytic Perspective

vs.

Psychoanalytic Therapy

originated by Sigmund Freud

The psychoanalytic approach digs into the hidden parts of our minds that we don't normally access but that still shape who we are and what we do.

In psychoanalytic therapy, a therapist might use techniques like free association (saying whatever comes to mind) or dream analysis to help bring these unconscious thoughts into awareness so the person can understand and resolve them.

5. The Subjectivist Perspective

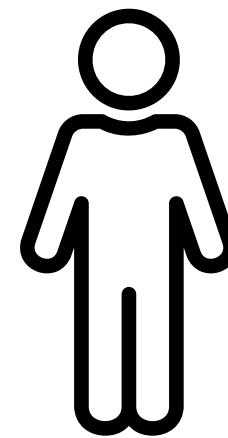
- The subjectivist perspective contends that human behavior is a function of the perceived world, not the objective world.
- To understand human social behavior, this view holds, we must grasp the person's own 'definition of the situation', which is expected to vary by culture, personal history, and current motivational state. This perspective, then, is the most open to cultural and individual differences and to the effects of motivation and emotion.

Key Ideas:

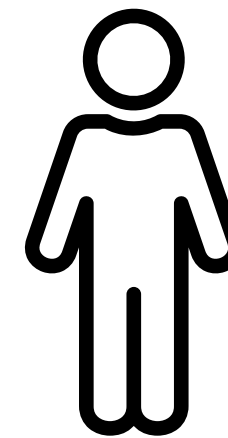
- Reality is viewed differently by each person
- How you interpret events matters more than the events themselves
- Personal meaning is crucial to understanding behavior
- There's no single "objective truth" - just different ways people make sense of things
- People actively create meaning rather than just responding to the world

Imagine three people witnessing the same car accident:

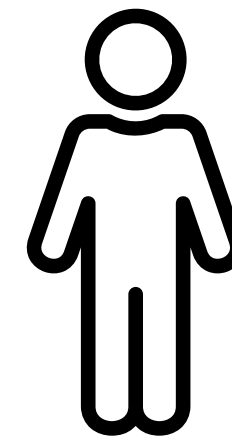
- Person A sees it as a terrible tragedy and feels deeply upset
- Person B sees it as confirmation that people drive too fast in that area
- Person C, who recently survived a similar accident, has a panic attack



A



B



C

Thank You

Nolen-Hoeksema Fredrickson Loftus Wagenaar

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